Zürcher Hochschule
für Angewandte Wissenschaften

Life Sciences und
Facility Management

Imaging for the Life Sciences

Master of Science in Life Sciences
CO4
Cluster Computation

Basics

A short history of images

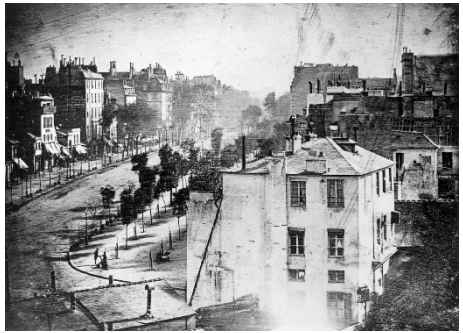
- Humans generate, record and transmit images to store and pass on information since the early days

Painting of a bison from the Altamira cave, dating from around 36,000 years ago. (Source: Wikimedia)

A short history of images

- Humans generate, record and transmit images to store and pass on information since the early days
- First photograph was pioneered in the 1820s by the Frenchmen Joseph Nicéphore Niépce and Louis Daguerre

1826: View from the Window at Le Gras by Niépce. Possibly the earliest surviving photograph. Source



1838: Boulevard du Temple, Paris. Photography by Louis Daguerre. Source

A short history of images

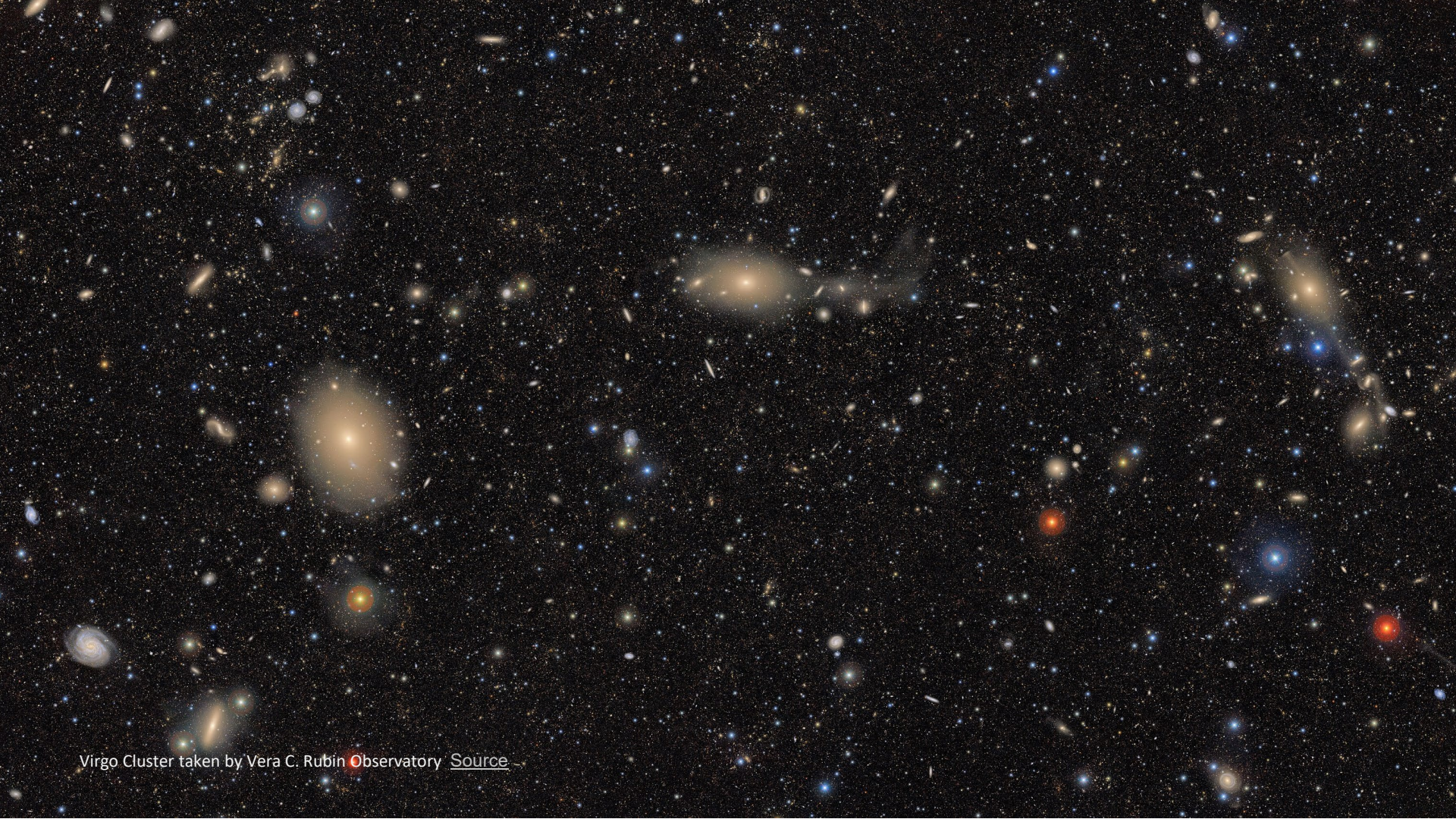
- Humans generate, record and transmit images to store and pass on information since the early days
- First photograph was pioneered in the 1820s by the Frenchmen Joseph Nicéphore Niépce and Louis Daguerre
- First digital image was taken by Russel Kirsch in 1957 using a scanner
- First digital camera using a CCD (charge-coupled device) sensor from Willard S. Boyle and George E. Smith in 1969

1957: First digital image ever created, by Russel Kirsch. Source

A short history of images

- Humans generate, record and transmit images to store and pass on information since the early days
- First photograph was pioneered in the 1820s by the Frenchmen Joseph Nicéphore Niépce and Louis Daguerre
- First digital image was taken by Russel Kirsch in 1957 using a scanner
- First digital camera using a CCD (charge-coupled device) sensor from Willard S. Boyle and George E. Smith in 1969
- The first images from the Vera C. Rubin Observatory's 3.2-gigapixel CCD camera, the largest camera ever built, in 2025

Vera C. Rubin Observatory (2023). Source

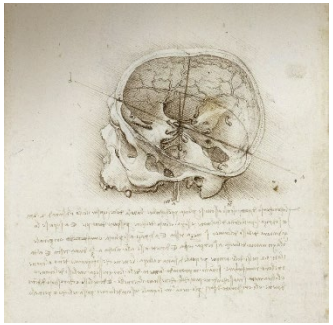


Virgo Cluster taken by Vera C. Rubin Observatory Source

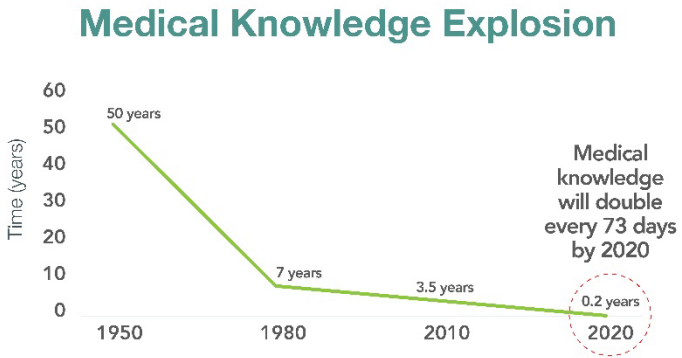
Medical Imaging

Evolution of Medical Imaging

- **Ancient Beginnings:** Early civilizations used basic observation and touch and feel
- **Renaissance to Modern Era:** Leonardo da Vinci's drawings
- **Imaging:** Microscope invented around 1590 by Hans Janssen and Zacharias Janssen
- **Radiography:** Röntgen's discovery of X-rays revolutionized diagnosis with X-ray imaging
- **Ultrasound:** Mid-20th century development for versatile imaging
- **CT and MRI:** Detailed cross-sectional and soft tissue imaging since the 1970s
- **Nuclear Medicine and PET:** Functional insights from radioactivity discovery
- **Future:** AI integration, molecular imaging, collaborative innovation



Glimpse into the left side of a human skull, about 1651
© Royal Collection Trust / Her Majesty Queen Elizabeth II 2013

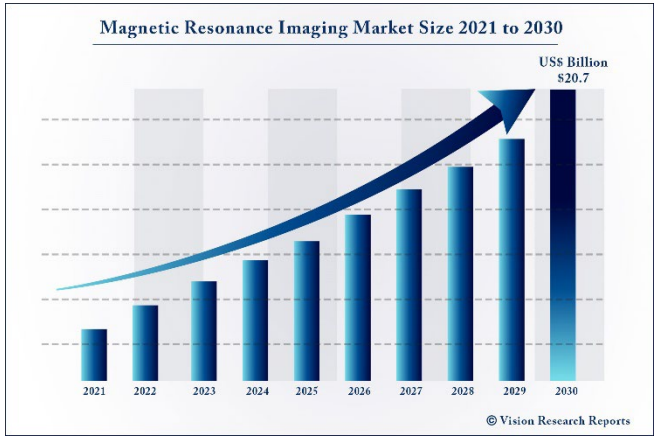


Graphic source, NCBI, "CHALLENGES AND OPPORTUNITIES FACING MEDICAL EDUCATION" Peter Densen, MD, 2011

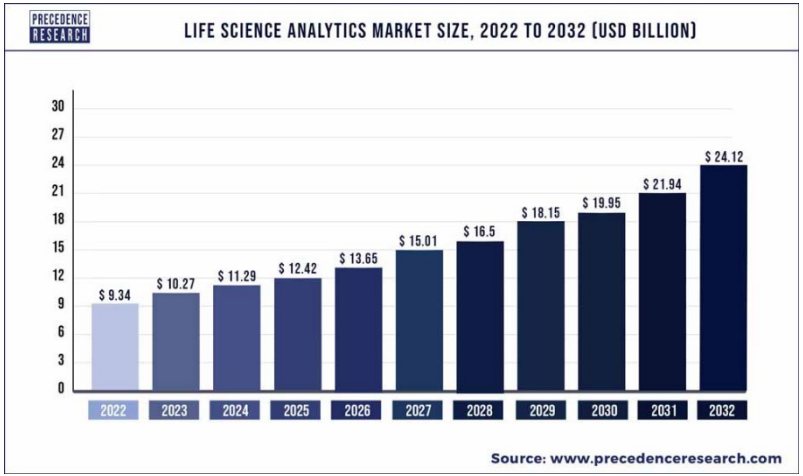
Imaging for the Life Sciences

… can provide more than just anatomical information.

The amount of acquired medical images is constantly increasing.



https://www.visionresearchreports.com/magnetic-resonance-imaging-market/38818



https://www.precedenceresearch.com/life-science-analytics-market

Imaging for the Life Sciences

Direct or indirect imaging

Image processing

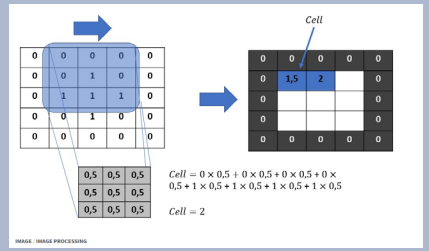


Image representation

Perception of images

Imaging for the Life Sciences

Direct or indirect imaging

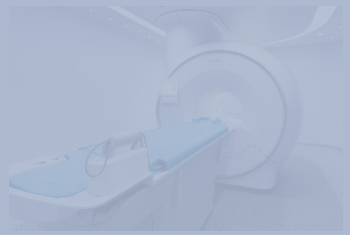


Image processing

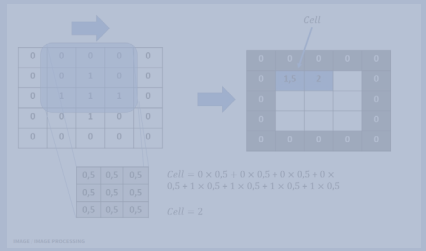
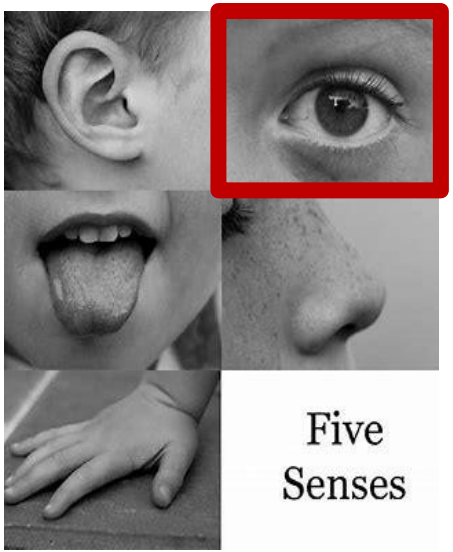


Image representation

Perception of images

Perception of images

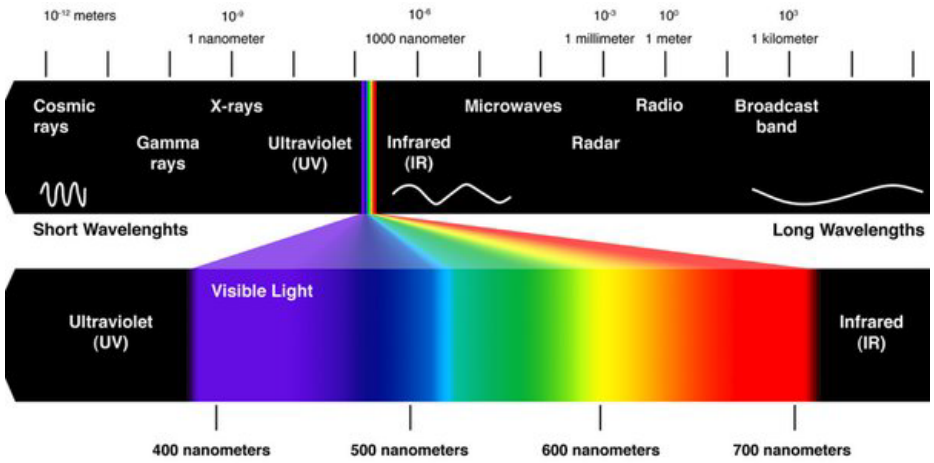


More than 99% of the activity of the human brain is involved in processing images from the visual cortex.

A visual image is rich in information.

Confucius said:
"**A picture is worth a thousand words**".

Perception of images



https://www.quora.com/What-is-visible-light-and-how-much-of-the-spectrum-does-it-cover-compared-to-the-others

Imaging for the Life Sciences

Direct or indirect imaging

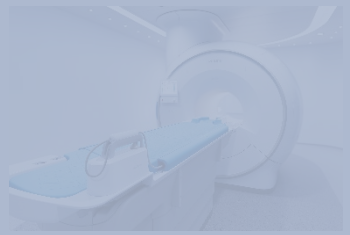


Image processing

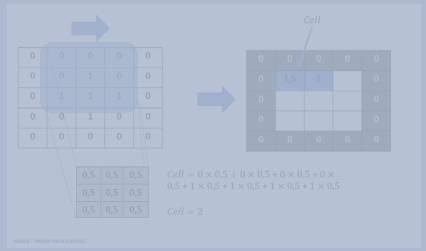


Image representation

Perception of images

Imaging for the Life Sciences

Direct or indirect imaging

Image processing

Image representation

Perception of images

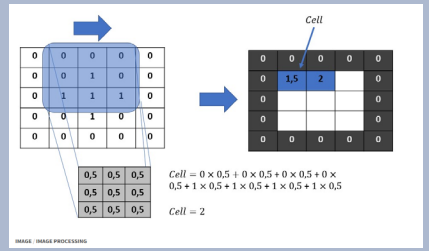


Image Processing

- Image processing is the manipulation and analysis of digital images to extract meaningful information or enhance their visual appearance.

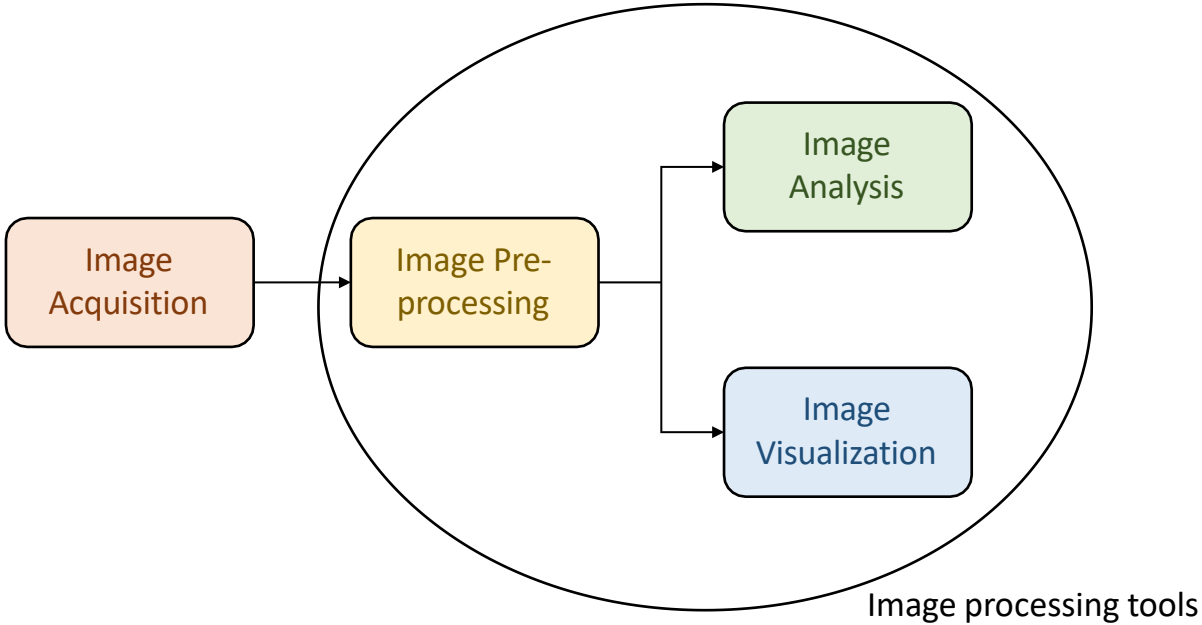
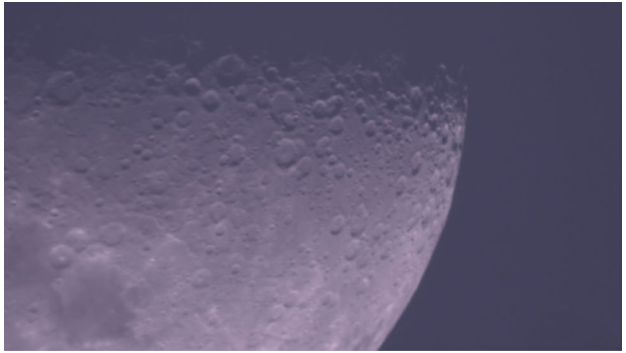
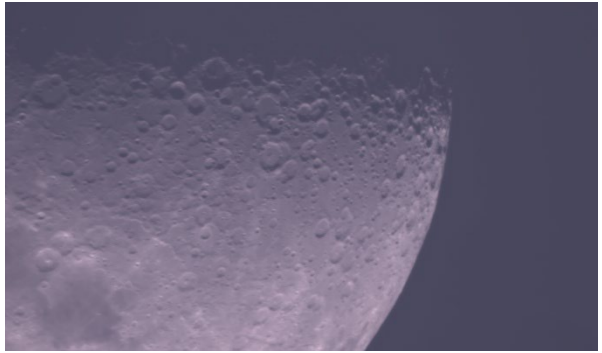


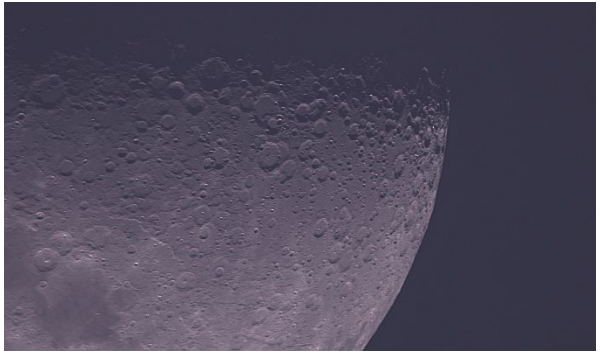
Image Processing: Example



Acquisition: Video 60fps



after image stacking



after deconvolution/sharpening

Imaging for the Life Sciences

Direct or indirect imaging

Image processing

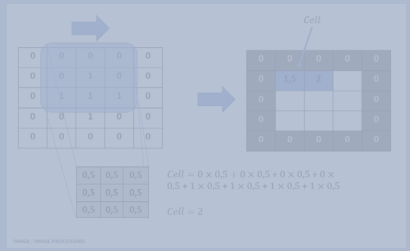
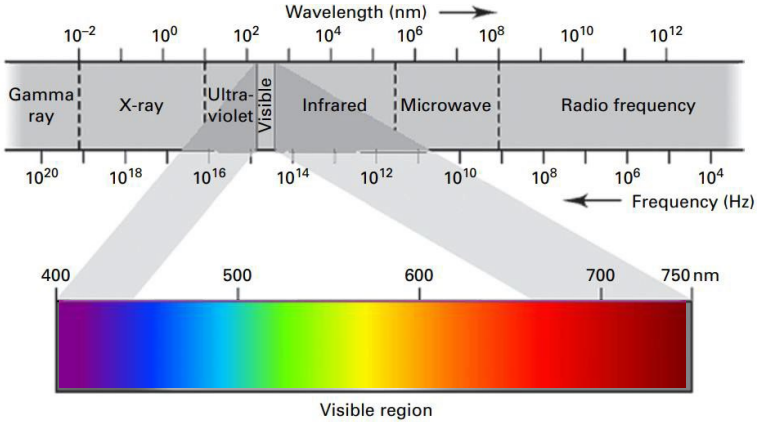


Image representation

Perception of images

Direct imaging

Classification of imaging by type of radiation or field used



The electromagnetic spectrum arranged according to the energy of the photons, or the frequency of the waves.

Direct Imaging:

- Direct imaging refers to a method where the detected signal directly corresponds to the phenomenon or object being imaged.
- The detected signal comes directly from the target object without any intermediate steps or processes.

Direct imaging

Imaging systems

Direct Imaging:

- Examples of direct imaging include photography, where light from the object directly exposes the film or sensor to create an image, and direct observation through a microscope or telescope.

Indirect imaging

Three different types of waves can be used:

1. Particle Waves
   - Particle waves exhibit wave-particle duality, where particles like electrons display both particle-like and wave-like behavior.
   - Characterized by their associated wavelengths, particle waves have much smaller wavelengths compared to electromagnetic waves.
   - Example: Electron waves are utilized in electron microscopes for high-resolution imaging.

2. Electromagnetic (EM) Waves
   - EM waves are oscillating electric and magnetic fields that propagate through space.
   - These waves have a wide range of wavelengths, from radio waves to gamma rays.
   - Examples: Radio waves for communication, visible light for vision, and X-rays for medical imaging.

3. Pressure Waves
   - Pressure waves, also known as mechanical or sound waves, are fluctuations in pressure that propagate through a medium, such as air or water.
   - Characterized by their relatively longer wavelengths compared to particle waves and EM waves.
   - Example: Sound waves are utilized in various applications, including communication, sonar, and medical imaging.

Indirect imaging

Comparison

Features	Particle Waves	Waves EM	Pressure Waves
Wave-particle duality	✓	X	X
Wavelengths	Short	Variable	Long
Examples	Electron waves	Radio waves	Sound waves
Applications	High-resolution imaging in electron microscopy	Communication, imaging (e.g., MRI, X-ray),	medical ultrasound

Imaging Methods

Different medical imaging modalities provide complementary information.

Microscopy

1. Soft X-ray Tomography (SXT)
2. High Resolution Episcopic Microscopy (HREM)
3. Electron Microscopy (EM)
    - Transmission Electron Microscopy (TEM)
    - Scanning Electron Microscopy (SEM)
4. Light Sheet Fluorescence Microscopy (LSFM)
5. MultiPhoton Microscopy (MPM)
6. Confocal Microscopy (CCM)
7. Atomic Force Microscopy (AFM)
8. Super resolution Microscopy (e.g., STORM, PALM, STED)
9. Fluorescence Lifetime Imaging Microscopy (FLIM)

Medical Imaging

10. Computed Tomography (CT)
11. Magnetic Resonance Imaging (MRI)
    - Diffusion Tensor Imaging (DTI)
    - Functional Magnetic Resonance Imaging (fMRI)
    - Magnetic Resonance Angiography (MRA)
    - Magnetic Resonance Spectroscopy (MRS)
    - Magnetic Resonance Elastography (MRE)
12. Positron Emission Tomography (PET)
    - Positron Emission Tomography - Computed Tomography (PET-CT)
    - Positron Emission Tomography - Magnetic Resonance Imaging (PET-MRI)
13. Single Photon Emission Computed Tomography (SPECT)
14. Photoacoustic Imaging (PAI)
15. Optical Coherence Tomography (OCT)
16. X-ray Radiography
17. Cryo-Electron Microscopy (Cryo-EM)

Optical Imaging

18. Optical Imaging including bioluminescence and fluorescence imaging (OI)
19. Mass Spec Imaging (MSI)
20. Raman Imaging

Medical Diagnostics

21. Ultrasound (US)
22. Functional Near-Infrared Spectroscopy (fNIRS)
23. Electroencephalography (EEG)
24. Electromyography (EMG)
25. Electrocardiography (ECG)

Endoscopic Imaging

26. Endoscopy (ES)

Surgical Procedures

27. Biopsy (BX)
28. Laparoscopy (LAP)

Imaging Methods

Different medical imaging modalities provide complementary information.

Microscopy

1. Soft X-ray Tomography (SXT)
2. High Resolution Episcopic Microscopy (HREM)
3. Electron Microscopy (EM)
    - Transmission Electron Microscopy (TEM)
    - Scanning Electron Microscopy (SEM)
4. Light Sheet Fluorescence Microscopy (LSFM)
5. MultiPhoton Microscopy (MPM)
6. **Confocal Microscopy (CCM)**
7. Atomic Force Microscopy (AFM)
8. Super resolution Microscopy (e.g., STORM, PALM, STED)
9. **Fluorescence Lifetime Imaging Microscopy (FLIM)**

Medical Imaging

10. **Computed Tomography (CT)**
11. **Magnetic Resonance Imaging (MRI)**
    - **Diffusion Tensor Imaging (DTI)**
    - **Functional Magnetic Resonance Imaging (fMRI)**
    - **Magnetic Resonance Angiography (MRA)**
    - **Magnetic Resonance Spectroscopy (MRS)**
    - **Magnetic Resonance Elastography (MRE)**
12. **Positron Emission Tomography (PET)**
    - **Positron Emission Tomography - Computed Tomography (PET-CT)**
    - **Positron Emission Tomography - Magnetic Resonance Imaging (PET-MRI)**
13. **Single Photon Emission Computed Tomography (SPECT)**
14. **Photoacoustic Imaging (PAI)**
15. Optical Coherence Tomography (OCT)
16. **X-ray Radiography**
17. Cryo-Electron Microscopy (Cryo-EM)

Optical Imaging

18. **Optical Imaging including bioluminescence and fluorescence imaging (OI)**
19. Mass Spec Imaging (MSI)
20. Raman Imaging

Medical Diagnostics

21. **Ultrasound (US)**
22. Functional Near-Infrared Spectroscopy (fNIRS)
23. Electroencephalography (EEG)
24. Electromyography (EMG)
25. Electrocardiography (ECG)

Endoscopic Imaging

26. Endoscopy (ES)

Surgical Procedures

27. Biopsy (BX)
28. Laparoscopy (LAP)

Imaging Methods

Individual resp. group work

List of Participants: group work - CO4, Imaging for the Life Sciences

spring semester 26: We 16-22

last name	first name	school	Team
Aebli	Kai	ZHAW	1.Soft X-ray Tomography (SXT)
Ajegunma	Jamal	ZHAW	1.Soft X-ray Tomography (SXT)
Begemann	Jan	ZHAW	2.High Resolution Episcopic Microscopy (HREM)
Bützer	Julian	ZHAW	2.High Resolution Episcopic Microscopy (HREM)
Coronado Sanabria	Sara Julieth	HES-SO	3.Transmission Electron Microscopy (TEM)
da Silva do Aido	Luana	ZHAW	3.Transmission Electron Microscopy (TEM)
Davolio	Noemi	ZHAW	4.Scanning Electron Microscopy (SEM)
Ezekwu	Yannik	ZHAW	4.Scanning Electron Microscopy (SEM)
Filo	Alex	ZHAW	5.Light Sheet Fluorescence Microscopy (LSFM)
Giacomelli	Robin	ZHAW	5.Light Sheet Fluorescence Microscopy (LSFM)
Giora	Triantafyllia	ZHAW	6.MultiPhoton Microscopy (MPM)
Hertach	Oliver	ZHAW	6.MultiPhoton Microscopy (MPM)
Herzog	Benjamin	ZHAW	7.Atomic Force Microscopy (AFM)
Ho	Ka Men	ZHAW	7.Atomic Force Microscopy (AFM)
Hooja	Harshita	ZHAW	8.Super resolution Microscopy (e.g., STORM, PALM, STED)
Kobivasan	Kajenthini	ZHAW	8.Super resolution Microscopy (e.g., STORM, PALM, STED)
Li	Kun	ZHAW	9.Cryo-Electron Microscopy (Cryo-EM)
Morales	Maria	ZHAW	9.Cryo-Electron Microscopy (Cryo-EM)
Müller	Cédric	ZHAW	10.Optical Coherence Tomography (OCT)
Niklaus	Cyrille	ZHAW	10.Optical Coherence Tomography (OCT)
Nüesch	Fiona	ZHAW	11.Mass Spec Imaging (MSI)
Palese	Luigi	ZHAW	11.Mass Spec Imaging (MSI)
Pfister	Michèle	ZHAW	12.Raman Imaging
Rappaz	Stéphane	ZHAW	12.Raman Imaging
Ruppel	Paula	ZHAW	13. Functional Near Infrared
Schmid	Michelle	ZHAW	13. Functional Near Infrared
Siebeneicher	Rebecca	ZHAW	14.Endoscopy (ES)
Sürer	Serna	ZHAW	14.Endoscopy (ES)
Viganò	Giorgia	HES-SO	15.Biopsy (BX)
Vinzens	Livia	ZHAW	15.Biopsy (BX)
Zimmermann	Alex Ethan	FHNW	

Individual resp. group work

Task:

1. Please create one ppt slide per imaging modality (investigate method, collect important information, create ppt with 1-2 images)
   - The following information should be included:
     - ✓ How does it work?
     - ✓ Purpose?
     - ✓ Advantages?
     - ✓ Disadvantages?

   Time 30 min

2. Collect the slides and create a group presentation

   Time 5 min
   - Group 1: Team 1-7
   - Group 2: Team 8-15

3. Fill in values https://andreaszhaw.github.io/ImagingModalities/

4. Every team presents their method in a maximum of 2 min/method

Imaging Methods

Individual resp. group work

Individual resp. group work

Task:

1. Please create one ppt slide per imaging modality (investigate method, collect important information, create ppt with 1-2 images)
   - The following information should be included:
     - ✓ How does it work?
     - ✓ Purpose?
     - ✓ Advantages?
     - ✓ Disadvantages?

Time 30 min

2. Collect the slides and create a group presentation
   - Group 1: Team 1-7
   - Group 2: Team 8-15

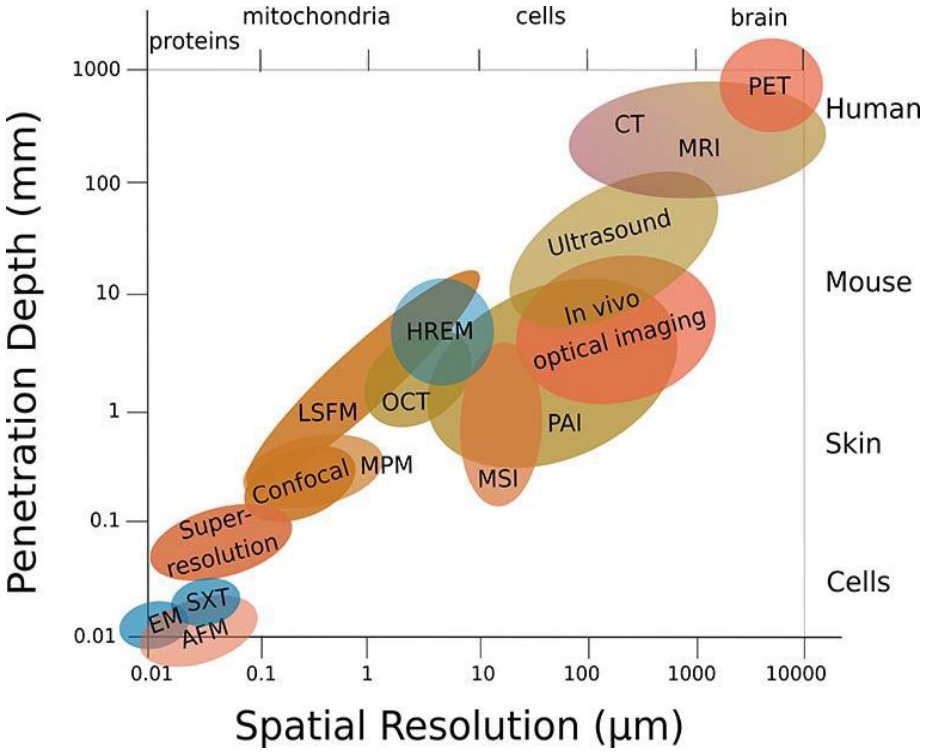
Time 5 min

3. Fill in values https://andreaszhaw.github.io/ImagingModalities/

4. Every team presents their method in a maximum of 2 min/method

Direct or indirect imaging

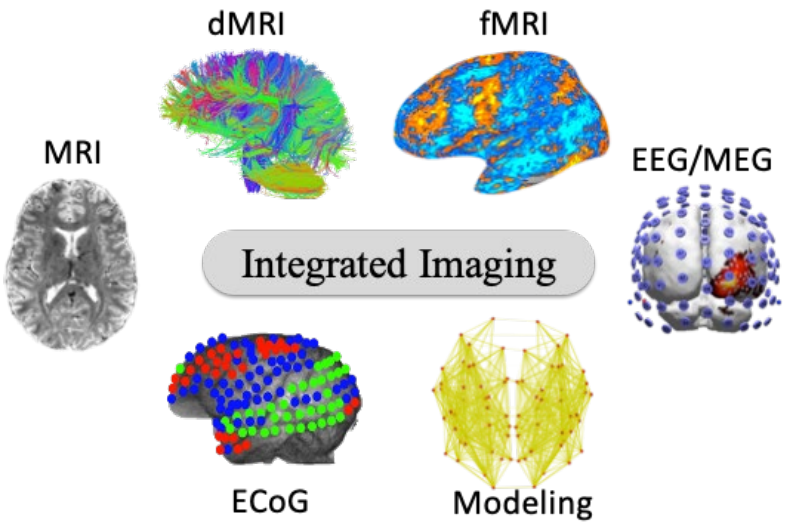
Different medical imaging modalities provide complementary information.

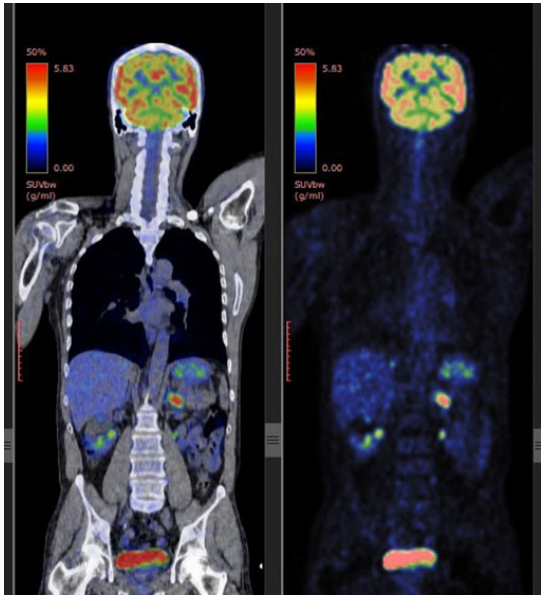


Holistic imaging accessing all relevant information spaces from structural to functional imaging. Representation for the modalities mentioned in the text: SXT, Soft X-ray Tomography; HREM, High Resolution Episcopic Microscopy; EM, Electron Microscopy; CT, Computed Tomography; OI, Optical Imaging including bioluminescence and fluorescence imaging; LSFM, Light Sheet Fluorescence Microscopy; MPM, MultiPhoton Microscopy, CCM, Confocal, Confocal Microscopy; MRI, Magnetic Resonance Imaging; US, Ultrasound, PAI, Photoacoustic Imaging; OCT, Optical Coherence Tomography; AFM, Atomic Force Microscopy; MSI, Mass Spec Imaging; Superresolution, Microscopy such as STORM, PALM, STED; PET/SPECT, Positron Emission Tomography/Single Photon Emission Computed Tomography.

Direct or indirect imaging

Multi-modal image analysis





Imaging for the Life Sciences

Direct or indirect imaging

Image processing

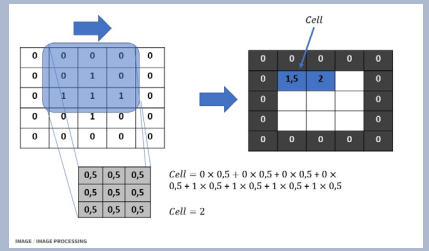


Image representation

Perception of images

hocn@zhaw.ch